

(1) Steady displacement from discrete plate simulation (δ_{plate}) and theoretical prediction (δ_{EB}), and their normalized difference.

After modeling, the mesh looks like this:

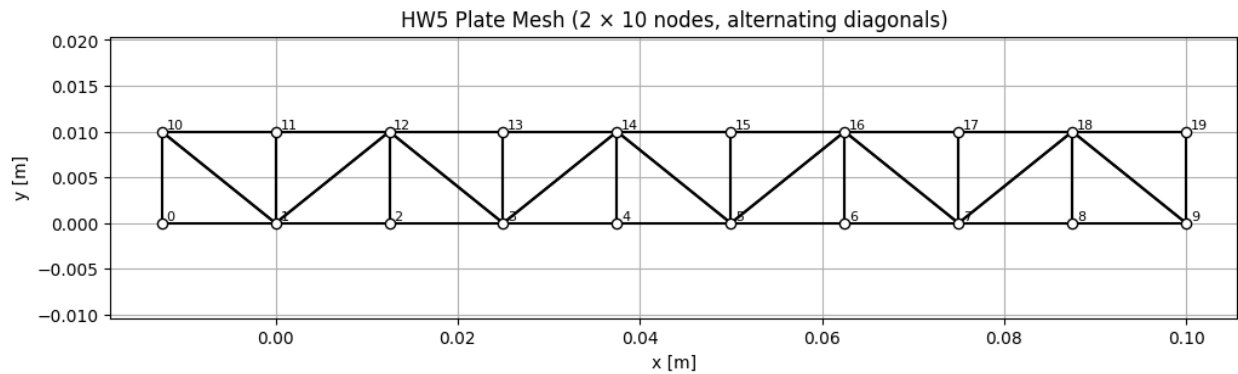


Figure 1. plot of Mesh with fixed nodes in the left side

After applying the structure's mass, running the code, and calculating the deflection analytically, we can determine both the simulated δ and the δ from the formula, shown below.

$$\delta_{plate} \text{ (steady)} = 0.03483798189574103 \text{ m}$$

$$\delta_{EB} \text{ (theory)} = 0.036787500000000001 \text{ m}$$

And by using $\frac{\delta_{EB} - \delta_{plate}}{\delta_{EB}} * 100$ we'll get:

$$\text{Normalized diff} = 5.3 \%$$

(2) A plot of δ_{plate} vs. Time t.

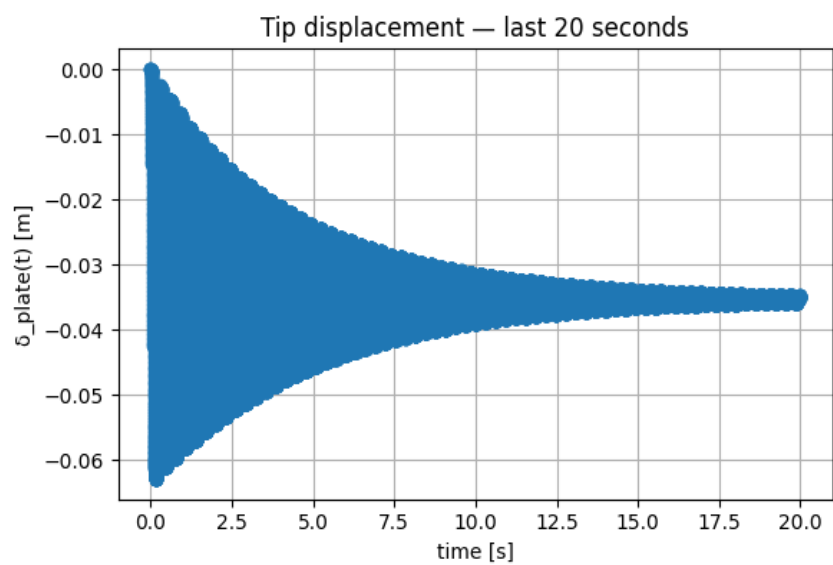


Figure 2. plot of δ_{plate} vs. Time t.

In Fig 2 we can see the plot of δ_{plate} vs. Time and we'll arrive at a steady state response after around 17.5 seconds with less than 2% fluctuation, but here I've taken the last value to be the final steady state deflection.